# Fifth Semester B. Tech. ( Electronics and Communication Engineering ) Examination

## WIRELESS COMMUNICATION

Time : 3 Hours ] [ Max. Marks : 60

**Instructions to Candidates :—**

(1) All questions carry marks as indicated.
(2) All questions are compulsory.
(3) Assume suitable data wherever necessary.
(4) Illustrate your answers with help of neat sketch wherever necessary.

1. (a) Demonstrate Frequency Reuse concept in wireless Communication System and making its use, illustrate relationship between cluster size M and system capacity C. 5(CO1)

   (b) A Mobile Communication system is allocated with RF spectrum of 25 MHz and uses RF channel Bandwidth of 25 KHz :

   (i) If the service area is divided into 20 cells with a frequency reuse pattern of 4, compute the system capacity.

   (ii) The cell size is reduced to the extent that the service area is now covered with 100 cells. Compute the system Capacity which is keeping the frequency reuse pattern as 4.

   (iii) Comment on the results. 5(CO4)

2. (a) Summarize the concept of multipath Fading with the help of diagram. 4(CO2)

   (b) Apply the concepts of Reflection, Refraction and Scattering to demonstrate signal propagation in Wireless Communication. 6(CO2)

3. (a) Illustrate the concept of GMSK with the help of diagram. 5(CO3)

   (b) Compare space Diversity and Frequency Diversity techniques. 5(CO4)

4. Clarify the concept of multiple access technique in CDMA and hence describe DS-CDMA and FH-CDMA. 10(CO3)

5. Describe in detail GSM channels which include both Traffic (TCHs) and control (CCHs) channels. 10(CO5)

6. (a) Does deployment of Non-Othogonal Multiple Access improves the system capacity. If yes then provide appropriate explanation to support your answer. 5(CO5)

   (b) Consider a Reliance Network cell with $S = 2$ channels with 100 Mobile stations generating call requests on an average 30 requests/hr. Average Holding time 360 seconds. Calculate load a, no of re-routed calls and efficiency of the system. (Blocking Probability $B = 0.53$). 5(CO4)